

● RAG Learning

3.3 Compound Measures & Rates of Change

AQA GCSE Mathematics 8300 — Ratio, Proportion & Rates of Change — spec refs R1, R11, R14, R15 — Papers 1 (non-calculator), 2 and 3 (calculator)

3.3.1 Converting Between Standard Units

Before you can use any compound measure you must have the quantities in **consistent units**. Many marks are lost simply because a time was left in minutes when the speed was in km per hour, or an area was in cm^2 when the pressure was wanted in N/m^2 .

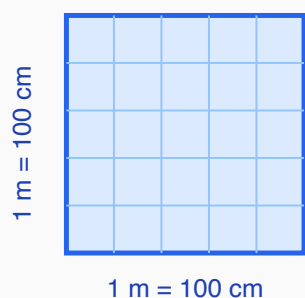
Length, mass, capacity and time

Quantity	Key facts
Length	$10 \text{ mm} = 1 \text{ cm}$ $100 \text{ cm} = 1 \text{ m}$ $1000 \text{ m} = 1 \text{ km}$
Mass	$1000 \text{ mg} = 1 \text{ g}$ $1000 \text{ g} = 1 \text{ kg}$ $1000 \text{ kg} = 1 \text{ tonne}$
Capacity	$1000 \text{ ml} = 1 \text{ litre}$ $100 \text{ cl} = 1 \text{ litre}$ $1 \text{ ml} = 1 \text{ cm}^3$ $1000 \text{ litres} = 1 \text{ m}^3$
Time	$60 \text{ s} = 1 \text{ min}$ $60 \text{ min} = 1 \text{ h}$ $3600 \text{ s} = 1 \text{ h}$ $24 \text{ h} = 1 \text{ day}$

Exam Tip: Time is **not** decimal. 2.25 hours is 2 hours 15 minutes ($0.25 \times 60 = 15$), not 2 hours 25 minutes. To change minutes into hours, divide by 60: 40 minutes = $\frac{40}{60} = \frac{2}{3}$ hour.

Area and volume units

Area and volume conversions use the **square** and **cube** of the length conversion factor, because every dimension is converted.



$$\begin{aligned}
 1 \text{ m}^2 & \\
 &= 100 \times 100 \\
 &= 10\,000 \text{ cm}^2
 \end{aligned}$$

Area: $1 \text{ cm}^2 = 100 \text{ mm}^2$ $1 \text{ m}^2 = 10\,000 \text{ cm}^2$

Volume: $1 \text{ cm}^3 = 1000 \text{ mm}^3$ $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$

If $1 \text{ m} = 100 \text{ cm}$, then $1 \text{ m}^2 = 100^2 \text{ cm}^2$ and $1 \text{ m}^3 = 100^3 \text{ cm}^3$.

Worked Example: Convert 4500 cm^3 into m^3 .

$1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$, so divide by 1 000 000:

$$4500 \div 1\,000\,000 = \mathbf{0.0045 \text{ m}^3}$$

Worked Example: Convert 3.2 m^2 into cm^2 .

$$3.2 \times 10\,000 = \mathbf{32\,000 \text{ cm}^2}$$

3.3.2 Compound Units

A **compound unit** combines two (or more) units to describe a rate — how much of one quantity there is *per* unit of another. Examples: km/h (kilometres per hour), m/s, £/hour (rate of pay), pence per gram (unit price), g/cm³ (density), N/m² (pressure).

The word “per” means “divide”. A rate of pay of £12 per hour means $\text{pay} \div \text{hours} = 12$. Read the compound unit and it tells you the formula: km/h means kilometres $\div$ hours.

Converting compound units

Convert each part of the unit separately.

Worked Example: Convert 72 km/h into m/s.

72 km = 72 000 m and 1 hour = 3600 s

$$72\,000 \div 3600 = \mathbf{20\text{ m/s}}$$

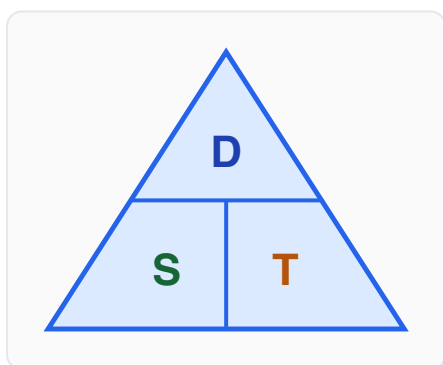
Shortcut: km/h $\rightarrow$ m/s divide by 3.6; m/s $\rightarrow$ km/h multiply by 3.6.

Worked Example: Convert a density of 2.7 g/cm³ into kg/m³.

1 m³ = 1 000 000 cm³, so 1 m³ has mass $2.7 \times 1\,000\,000 = 2\,700\,000\text{ g} = 2700\text{ kg}$

Density = **2700 kg/m³**

3.3.3 Speed, Distance and Time



$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Distance} = \text{Speed} \times \text{Time} \quad \text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

Cover the quantity you want in the triangle to see the formula.

Worked Example: A car travels for 45 minutes at 64 km/h. How far does it go?

45 minutes = 0.75 hours

$$\text{Distance} = 64 \times 0.75 = \mathbf{48\text{ km}}$$

Worked Example: A lorry travels 84 miles at an average speed of 56 mph. It leaves at 10:45. When does it arrive?

$$\text{Time} = 84 \div 56 = 1.5 \text{ hours} = 1 \text{ hour } 30 \text{ minutes}$$

$$\text{Arrival time} = 10:45 + 1:30 = \mathbf{12:15}$$

Average speed for a journey in several parts

Average speed is **total distance $\div$ total time**. It is *not* the mean of the separate speeds.

Worked Example: A car travels 30 km at 60 km/h and then 30 km at 40 km/h. Find the average speed.

$$\text{Time 1} = 30 \div 60 = 0.5 \text{ h} \quad \text{Time 2} = 30 \div 40 = 0.75 \text{ h}$$

$$\text{Average speed} = \frac{60}{1.25} = \mathbf{48 \text{ km/h}} \quad (\text{not } 50 \text{ km/h})$$

Exam Tip: Include any stops in the total time when asked for the average speed “for the whole journey”. Always state units, and check the answer is sensible (a person walks at about 5 km/h, a car on a motorway about 110 km/h).

3.3.4 Rates of Pay and Unit Pricing

Rates of pay (£ per hour) and unit prices (pence per gram, £ per litre) are compound measures used in everyday contexts. AQA expects you to **make comparisons** — for example deciding which pack is the best buy.

Worked Example: Pack A: 500 g for £1.80. Pack B: 750 g for £2.55. Which is better value?

$$\text{Pack A: } 180 \div 5 = 36\text{p per } 100 \text{ g}$$

$$\text{Pack B: } 255 \div 7.5 = 34\text{p per } 100 \text{ g}$$

Pack B is better value because it costs less per 100 g.

Worked Example: Amy earns £11.40 per hour. Ben earns £21 600 a year, working 40 hours a week for 48 weeks. Who has the higher hourly rate?

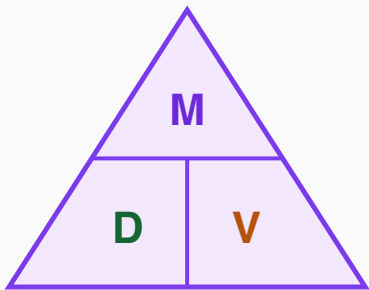
Ben's hours = $40 \times 48 = 1920$ Ben's rate = $21\,600 \div 1920 = \text{£}11.25$ per hour

Amy earns 15p per hour more.

Exam Tip: When comparing, compare **like with like** (cost per gram for every option, or amount per £1 for every option) and finish with a clear statement that uses your figures. A correct choice with no working scores nothing.

3.3.5 Density

Density is the mass per unit volume of a substance. Common units are g/cm^3 and kg/m^3 .



$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

$$\text{Mass} = \text{Density} \times \text{Volume} \quad \text{Volume} = \frac{\text{Mass}}{\text{Density}}$$

Worked Example: A block of wood measures 12 cm by 5 cm by 4 cm and has mass 168 g. Find its density.

$$\text{Volume} = 12 \times 5 \times 4 = 240 \text{ cm}^3$$

$$\text{Density} = 168 \div 240 = \mathbf{0.7 \text{ g/cm}^3}$$

Worked Example: 300 cm^3 of liquid A (density 1.2 g/cm^3) is mixed with 200 cm^3 of liquid B (density 0.9 g/cm^3). Find the density of the mixture.

$$\begin{aligned} \text{Mass of A} &= 300 \times 1.2 = 360 \text{ g} & \text{Mass of B} &= 200 \times 0.9 = 180 \text{ g} \\ \text{Density} &= \frac{360 + 180}{300 + 200} = \frac{540}{500} = \mathbf{1.08 \text{ g/cm}^3} \end{aligned}$$

Exam Tip: For mixtures, never average the two densities. Find the **total mass** and the **total volume**, then divide.

3.3.6 Pressure

Pressure is the force acting per unit area. Force is measured in newtons (N), so pressure is measured in N/m^2 (also called pascals, Pa) or N/cm^2 .

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} \quad \text{Force} = \text{Pressure} \times \text{Area} \quad \text{Area} = \frac{\text{Force}}{\text{Pressure}}$$

For the same force, a **smaller area gives a greater pressure**. A block resting on its smallest face exerts the greatest pressure.

Worked Example: A box of weight 72 N has a rectangular base 40 cm by 30 cm. Find the pressure on the floor in N/m^2 .

$$\text{Area} = 0.4 \times 0.3 = 0.12 \text{ m}^2$$

$$\text{Pressure} = 72 \div 0.12 = \mathbf{600 \text{ N/m}^2}$$

Worked Example: A cuboid 20 cm $\times$ 10 cm $\times$ 5 cm has weight 40 N. Find the greatest pressure it can exert.

Smallest face = $10 \times 5 = 50 \text{ cm}^2$

Greatest pressure = $40 \div 50 = \mathbf{0.8 \text{ N/cm}^2}$

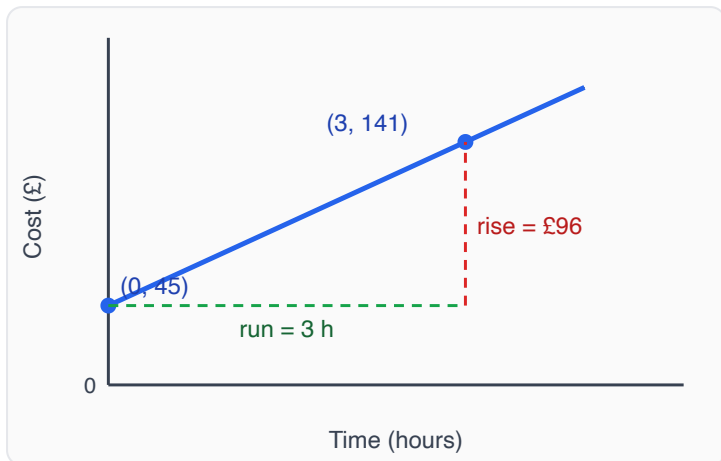
Exam Tip: Watch the area units. If the answer is wanted in N/m^2 , convert the lengths to metres *before* multiplying: $40 \text{ cm} \times 30 \text{ cm} = 0.4 \text{ m} \times 0.3 \text{ m} = 0.12 \text{ m}^2$ (not 1200 m^2).

3.3.7 Gradient of a Straight Line as a Rate of Change

When a graph is a straight line, its gradient tells you how fast the y -quantity changes for each unit of the x -quantity. The units of the gradient are " y -units per x -unit".

$$\text{Gradient} = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Graph	Gradient means	Units
Distance–time	Speed	m/s, km/h
Velocity–time	Acceleration	m/s^2
Cost–time (e.g. plumber)	Charge per hour	£ per hour
Depth of water–time	Rate of filling or emptying	cm per minute
Conversion graph (e.g. £ to €)	Exchange rate	€ per £



Worked Example: A plumber's charge is shown by a straight line through (0, 45) and (3, 141).

$$\text{Gradient} = \frac{141 - 45}{3 - 0} = \frac{96}{3} = 32 \rightarrow \text{the plumber charges } \mathbf{\pounds 32 \text{ per hour}}.$$

The y -intercept, 45, is a fixed **call-out charge of £45**.

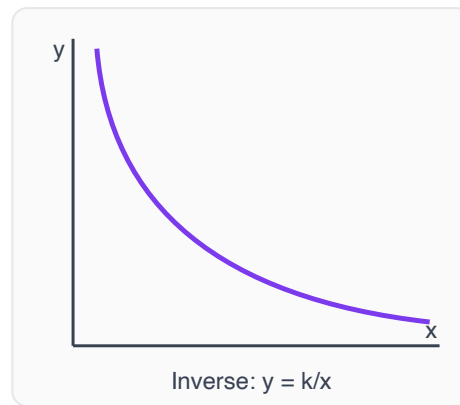
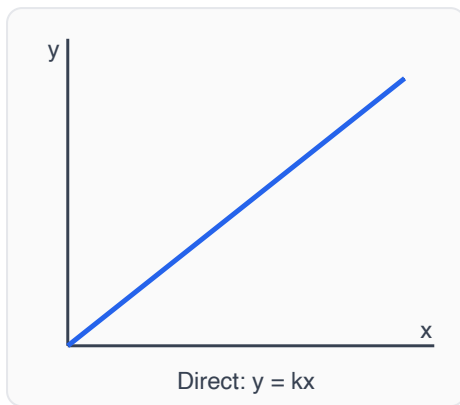
$$\text{Cost of a 5-hour job: } 32 \times 5 + 45 = \mathbf{\pounds 205}$$

Worked Example: The depth of water in a tank falls in a straight line from 80 cm at $t = 0$ to 20 cm at $t = 20$ minutes.

$$\text{Gradient} = \frac{20 - 80}{20} = -3 \rightarrow \text{the depth } \mathbf{\text{decreases by 3 cm per minute}}.$$

Exam Tip: A negative gradient means the quantity is decreasing. When asked to "interpret" a gradient, give the number *and* the units in context, e.g. "the water level falls by 3 cm every minute".

3.3.8 Graphs of Direct and Inverse Proportion



- **Direct proportion** ($y = kx$): a straight line **through the origin**. The gradient is the constant k (the rate). Doubling x doubles y .
- **Inverse proportion** ($y = \frac{k}{x}$): for $x > 0$ a curve that falls and gets closer and closer to both axes without touching them. Doubling x halves y , and $xy = k$ is constant.

Exam Tip: A straight line that does **not** pass through the origin (like the plumber's charge) shows a linear relationship but **not** direct proportion.

3.3.9 Average and Instantaneous Rates of Change

Higher

On a **curved** graph the rate of change is different at every point, so we need two ideas:

Chord: a straight line joining two points on a curve. Its gradient is the **average rate of change** between those two points.

Tangent: a straight line that touches the curve at one point (without crossing it there). Its gradient is the **instantaneous rate of change** at that point — this is the gradient of the curve at that point.



Average rate of change from $x = a$ to $x = b$ (gradient of chord):

$$\frac{f(b) - f(a)}{b - a}$$

Instantaneous rate of change at $x = a$: draw the tangent at $x = a$, choose two points far apart on it, and find rise $\div$ run.

Worked Example: Find the average rate of change of $y = x^2$ between $x = 1$ and $x = 4$.

$$\frac{4^2 - 1^2}{4 - 1} = \frac{16 - 1}{3} = 5$$

Estimating a gradient from a tangent

1. Place a ruler so it touches the curve at the point, with the curve the same distance away on each side.
2. Draw a long straight line (the tangent).
3. Pick two points on the tangent that are far apart and easy to read.
4. Gradient = change in $y \div$ change in x . Interpret it with units in context.

Worked Example: A car's distance–time graph is a curve. The tangent at $t = 5$ s passes through (2, 0) and (8, 54), with distance in metres.

$$\text{Gradient} = \frac{54 - 0}{8 - 2} = \frac{54}{6} = 9$$

The car's (instantaneous) speed at $t = 5$ s is about **9 m/s**.

Worked Example: The volume of water in a container is $V = 2t^2$ litres after t minutes.

Average rate from $t = 1$ to $t = 3$: $\frac{18 - 2}{3 - 1} = 8$ litres per minute.

The tangent at $t = 2$ passes through (1, 0) and (3, 16): gradient = $\frac{16}{2} = 8$ litres per minute.

So the instantaneous rate at $t = 2$ is **8 litres per minute**.

What the gradient of a curve tells you

Curved graph	Gradient of chord	Gradient of tangent
Distance–time	Average speed over a time interval	Speed at an instant
Velocity–time	Average acceleration	Acceleration at an instant
Volume–time	Average rate of flow	Rate of flow at an instant
Population–time	Average growth rate	Growth rate at an instant

If a distance–time curve gets steeper, the object is **accelerating** (speed increasing); if it gets less steep, the object is **decelerating**. A horizontal tangent means the rate of change is zero at that instant (e.g. a ball at its highest point).

Exam Tip: Gradients from a hand-drawn tangent are **estimates**, so mark schemes accept a range of answers. Draw the tangent long, use points far apart, and always say whether your value is an average rate (chord) or an instantaneous rate (tangent).

Summary

Measure	Formula	Typical units
Speed	distance $\div$ time	m/s, km/h, mph
Density	mass $\div$ volume	g/cm ³ , kg/m ³
Pressure	force $\div$ area	N/m ² (Pa), N/cm ²
Rate of pay	pay $\div$ hours	£ per hour
Unit price	cost $\div$ quantity	p per g, £ per litre
Rate of change (line)	gradient = rise $\div$ run	y -units per x -unit